VIGNESH RAVI

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Profile summary

Analytical and detail-oriented Data Analyst with hands-on experience in Python, SQL, data visualization, and machine learning. Proven track record in automating data workflows, analyzing datasets, and delivering actionable insights. Adept at using statistical methods, building dashboards, and creating predictive models to support data-driven decisions. Seeking to leverage analytical skills and technical expertise in a data-driven organization.

Work History

Research and Development Assistant, 09/2023 – 01/2025

Ezbon Global Sourcing – Chennai, India.

- Programmed and tested BLDC motor controllers tailored to customer requirements.
- Validated new and existing products by developing test methods and analyzing results.
- Created and maintained detailed Bills of Materials (BOMs).
- Debugged embedded systems to enhance product reliability and performance.
- Reduced design flaws by leading technical reviews and cross-functional collaboration.
- Coordinated project milestones and integrated customer feedback into development cycles.

Data Analyst Intern, 12/2022 – 05/2023

Microbiological Laboratory – Coimbatore, India.

- Developed an AI framework to interpret PCR data using ML/DL techniques, reducing reporting time significantly.
- Built a classification system using HRM data to improve disease detection accuracy.
- Applied CNNs to preprocess and classify peaks, improving signal clarity.
- Designed a GUI tool for automating data extraction and storage using relational databases.
- Developed a web-based reporting system with real-time visualization and feature extraction.
- Reduced PCR analysis time from hours to minutes, increasing lab throughput.

Project Portfolio

PyHRM

- Created a Python library for extracting and visualizing High-Resolution Melting (HRM) features such as peak areas, melting temps, and prominences
- Integrated interactive data visualizations and implemented image-based filtering techniques to remove noise, ensuring high-quality data for downstream analysis.
- Enabled accurate and automated feature extraction, improving efficiency in molecular diagnostic workflows.

Extractor

- Created *EXTRACTOR*, a desktop tool to automate the extraction and transformation of HRM raw data (.rex files) from Qiagen Rotor-Gene Q cyclers.
- Converted raw files into structured Excel outputs compatible with Q-Rex software, significantly reducing manual effort in data handling.
- Designed for high-throughput lab environments to streamline repetitive data entry and support internal analytics with clean, usable formats.

Skills

- **Data Analytics & Reporting [** Excel, SQL, Tableau, Power BI, Pandas, Seaborn]
- **Data Analysis & Cleaning [** Numpy, Pandas, Data Wrangling, Data Cleaning**]**
- **Data Visualization [** Matplotlib, Plotly, Seaborn**]**
- **Databases [** SQL, PostgreSQL**]**
- **Machine Learning & Statistics [** scikit-learn, SciPy, Regression/classification models, hypothesis testing]
- **Programming Languages [** Python, R, C++**]**
- **Version Control & Collaboration [** Git, GitHub**]**

Achievements

- Smart India Hackathon (SIH) – Participant
- NPTEL - Database Management Systems
- UiPath - RPA & Automation foundations

Education

- **Master of science:** Data analytics, 07/2021 – 05/2023

  Bharathiar University – Coimbatore, India – Distinction with first class
- **Bachelor of science:** Electronic Science, 06/2018 – 05/2021

  A.V.C College (Autonomous) – Mayiladuthurai, India – Gold medalist

Personal Details

Date of Birth : 06/04/2000

Nationality : Indian

Marital Status : Single

Languages : English (Read, Write, Speak),
Tamil (Read, Write, Speak).

Declaration

I hereby assure that the information furnished above are true to the best of my knowledge.